

Pharmaceutical Calculations 1- Solutions

Conversions

1. Convert the following into the given unit

a. 7250 mg

b. 44.4 mcg

c. 93.4 l

d. 65 mcg

e. 234 mcg

f. 0.25 kg

g. $45 \times 10^9 = 4.5 \times 10^{10}$ ng or 45000000000 ng

h. $123 \times 10^{-6} = 1.23 \times 10^{-4}$ g or 0.000123 g

Recipes

2. You have the following formula to create 250 ml of something medical sounding:

Ingredient A: 25 g

Ingredient B: 125 g

Ingredient C: 10 ml

Water to 250 ml

a. $\frac{750}{250} = 3$

Ingredient A: $25 \times 3 = 75$ g

Ingredient B: $125 \times 3 = 375$ g

Ingredient C: $10 \times 3 = 30$ ml

Water to 750 ml

b. 2 l = 2000 ml

$$\frac{2000}{250} = 8$$

Ingredient A: 200 g

Ingredient B: 1000 g

Ingredient C: 80 ml

Water to 2 l

c. $\frac{50}{250} = \frac{1}{5}$

Ingredient A: 5 g

Ingredient B: 25 g

Ingredient C: 2 ml

Water to 50 ml

Concentrations

3. 20 mg : 1 ml

200 mg : 10 ml

1000 mg : 50 ml

4.

5 mg : 1 ml

5000 mg : 1000 ml

5 g : 1 l

5. 40 g : 100 g

400 g : 1000 g

400 g : 1 kg

6. 2 mg : 1 ml

2000 mcg : 1 ml

200 mcg : 0.1 ml

800 mcg : **0.4 ml**

7. 20 g : 100 ml

20000 mg : 100 ml

200 mg : 1 ml

400 mg : 2 ml

50 mg : 0.25 ml

450 mg : **2.25 ml**

8. 8 g : 100 ml

8000 mg : 100 ml

800 mg : 10 ml

400 mg : 5 ml

1200 mg : 15 ml

9. 10% w/v means 10 g / 100 ml

10 g : 100 ml

20 g : 200 ml

5 g : 50 ml

25 g : 250 ml

10. 5 g : 100 ml

5000 mg : 100 ml

500 mg : 10 ml

250 mg : 5 ml

11. 15 % v/v means 15 ml / 100 ml

15 ml : 100 ml

1.5 ml : 10 ml

6 ml : 40 ml

6 ml active means $40 - 6 = \mathbf{34\text{ ml}}$ non-active

12. 0.2 g : 100 ml

2000 mg : 100 ml

20 mg : **1 ml**

13. $50 \times 40 = 2000$ mcg required by patient, or 2 mg

2 g : 100 ml

2000 mg : 100 ml

2 mg : **0.1 ml**

14. $80 \times 500 = 40000$ mcg required by patient, or 40 mg

5 g : 100 ml

5000 mg : 100 ml

1000 mg : 20 ml

10 mg : 0.2 ml

40 mg : **0.8 ml**

15. $50 \times 500 = 25000$ ng required by patient, or 25 mg

5 mcg : 1 ml

25 mcg : **5 ml**

16. $50 \times 2 = 100$ mg per minute required by patient

$100 \times 60 = 6000$ mg per 60 minutes (1 hour) required by patient (or 6 g)

1 g : 100 ml

6 g : **600 ml**

17. $5000 \times 70 = 350000 \text{ ng} = 350 \text{ mcg per minute}$

$350 \times 60 = 21000 \text{ mcg per hour}$

$2100 \times 10 = 210000 \text{ mcg per 10 hours (or 210 mg)}$

$2 \text{ g} : 100 \text{ ml}$

$2000 \text{ mg} : 100 \text{ ml}$

$200 \text{ mg} : 10 \text{ ml}$

$10 \text{ mg} : 0.5 \text{ ml}$

$210 \text{ mg} : \mathbf{10.5 \text{ ml}}$